

When Does Shift-Invariance Permit Adversarial Robustness?

A margin-preservation theory: proofs, open problems, and directions
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Abstract

This document collects, in full and with proofs, the current theory for when an emergent shift-invariance and ℓ_p adversarial robustness can co-exist in a classifier. It is written for external review: every claim is labelled by status (proved / verified numerically / empirical-only / conjecture / open / *retracted*). The established part is structural: the achievable robustness of a shift-invariant classifier is governed by the margin of the data in the invariant feature space divided by the Lipschitz constant of that feature, η/L (Section 2); this recovers the collapse of Ge et al. [3] exactly and reconciles Kamath et al. [4] and Tramèr et al. [7]. The optimization part — whether gradient descent *reaches* the robust invariant solution — is **open**; an earlier “architecture re-points the implicit bias” claim was found wrong and is retracted (Section 3), but the failure is informative and points to a concrete next data model. Section 4 lists conjectures, open questions, and directions, with the hints the experiments currently give. Section 5 is an overclaim audit and Section 6 lists what I would most like a reviewer to check or prove.

Contents

1	Setting and preliminaries	1
2	Structural theory: when invariance permits robustness	2
2.1	The exact invariant linear margin	2
2.2	The co-existence/trade-off dichotomy and the η/L criterion	3
2.3	Nonlinear escape: one convolution-plus-pooling layer computes the power spectrum	3
2.4	Recoveries and reconciliations	4
3	Optimization: does gradient descent reach the robust invariant solution?	5
4	Open problems, conjectures, and directions (with empirical hints)	6
5	Overclaim audit (status of every claim)	7
6	What I would most like a reviewer to check or prove	7

1 Setting and preliminaries

We use one-dimensional signals $x \in \mathbb{R}^d$. Everything extends verbatim to 2D images by replacing the cyclic group $\mathbb{Z}_d$ with $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$ and the cyclic DFT by its 2D analogue; the cyclic, 1D case is where

Ge et al. [3] and Kamath et al. [4] both work, so we keep it.

The cyclic group $G = \mathbb{Z}_d$ acts by the circular shift $(S_s x)_i = x_{(i-s) \bmod d}$. Each $S_s \in \mathbb{R}^{d \times d}$ is an orthogonal permutation, $S_s^\top = S_s^{-1} = S_{-s}$, and $S_a S_b = S_{a+b}$. The *orbit* of x is $\mathcal{O}(x) = \{S_s x : s \in G\}$. A class support is *orbit-closed* if it is a union of orbits.

Definition 1 (Invariant and AC subspaces). $V_{\text{inv}} = \{v \in \mathbb{R}^d : S_s v = v \ \forall s \in G\}$ and $V_{\text{ac}} = V_{\text{inv}}^\perp$. Write Π_{inv} for the orthogonal projector onto V_{inv} and $f_{\text{dc}}(x) = \frac{1}{\sqrt{d}} \sum_i x_i$ for the unit “DC” functional ($\|\frac{1}{\sqrt{d}} \mathbf{1}\|_2 = 1$).

Definition 2 (Robustness, consistency). For a score $f : \mathbb{R}^d \rightarrow \mathbb{R}$ correct at (x, y) ($y f(x) > 0$), the ℓ_p *robust radius* is $r_p(f, x) = \inf\{\|\delta\|_p : \text{sign}f(x + \delta) \neq y\}$. For a linear score $f(x) = w^\top x + b$, $r_p(f, x) = y f(x) / \|w\|_q$ with $1/p + 1/q = 1$. The *shift-consistency* is $\text{SC}(f) = \Pr_{x \sim \mathcal{D}, s \sim \text{Unif}(G)}[\text{sign}f(S_s x) = \text{sign}f(x)]$; an exactly invariant f has $\text{SC} = 1$.

Lemma 1 (Invariant subspace is the DC line). $V_{\text{inv}} = \text{span}(\mathbf{1})$, $\Pi_{\text{inv}} = \frac{1}{d} \mathbf{1} \mathbf{1}^\top$ (so $(\Pi_{\text{inv}} x)_i = \bar{x} := \frac{1}{d} \sum_j x_j$), and $V_{\text{ac}} = \{v : \sum_i v_i = 0\}$. The shifts are simultaneously diagonalized by the unitary DFT F ($F S_s F^* = \text{diag}(\omega^{sk})_k$, $\omega = e^{2\pi i/d}$); a shift multiplies Fourier coefficient k by a unit-modulus phase, so the power-spectrum coordinates $|\hat{x}_k|^2$ are shift-invariant, with $k = 0$ the DC line and $k \neq 0$ spanning V_{ac} .

Proof. If $S_s v = v$ for all s then in particular $S_1 v = v$ gives $v_i = v_{i-1}$ for all i , so v is constant, $v \in \text{span}(\mathbf{1})$; conversely $S_s \mathbf{1} = \mathbf{1}$. The projector onto a line $\text{span}(\mathbf{1})$ is $\mathbf{1} \mathbf{1}^\top / \|\mathbf{1}\|^2 = \frac{1}{d} \mathbf{1} \mathbf{1}^\top$. Its complement is $\{v : \mathbf{1}^\top v = 0\}$. Diagonalization by F is the standard fact that circulant matrices (the shifts are circulant permutations) are diagonalized by the DFT; S_s has eigenvalues ω^{sk} on the Fourier basis ϕ_k , and $|\widehat{S_s x}_k| = |\omega^{sk} \hat{x}_k| = |\hat{x}_k|$. *Status: proved.* $\square$

2 Structural theory: when invariance permits robustness

2.1 The exact invariant linear margin

Theorem 1 (Invariant linear margin = projection separation). A linear score $g(x) = w^\top x + b$ satisfies $g(S_s x) = g(x)$ for all x, s if and only if $w \in V_{\text{inv}} = \text{span}(\mathbf{1})$. Consequently, for orbit-closed classes X_+, X_- that are invariant-linearly separable, the maximum ℓ_2 margin achievable by an invariant linear classifier is exactly

$$\gamma_{\text{inv}} = \frac{1}{2} \left(\min_{x \in X_+} f_{\text{dc}}(x) - \max_{x \in X_-} f_{\text{dc}}(x) \right)_+ = \text{Sep}(\Pi_{\text{inv}} X), \quad (1)$$

attained by $w = \frac{1}{\sqrt{d}} \mathbf{1}$ (up to scale). Here $\text{Sep}(\cdot)$ is the hard-margin (half the gap between the projected classes) and $(\cdot)_+ = \max(\cdot, 0)$. The identity holds for any finite separable orbit-closed dataset, not only the single-image case of Ge et al. [3].

Proof. Invariance $\Leftrightarrow w \in V_{\text{inv}}$. $g(S_s x) = g(x)$ for all x means $w^\top S_s x = w^\top x$ for all x , i.e. $(S_s^\top w - w)^\top x = 0$ for all x , i.e. $S_{-s} w = w$. Requiring this for all s gives $w \in V_{\text{inv}} = \text{span}(\mathbf{1})$ by Lemma 1; conversely $w = c \mathbf{1}$ gives $w^\top S_s x = c \mathbf{1}^\top S_s x = c \mathbf{1}^\top x = w^\top x$ since S_s permutes coordinates.

Margin. Write $w = c \mathbf{1}$, $c \neq 0$. Then $w^\top x = c \sum_i x_i = c \sqrt{d} f_{\text{dc}}(x)$ and $\|w\|_2 = |c| \sqrt{d}$. For a correctly classified point the geometric margin is

$$\frac{y(w^\top x + b)}{\|w\|_2} = \frac{y(c \sqrt{d} f_{\text{dc}}(x) + b)}{|c| \sqrt{d}} = y(\text{sign}(c) f_{\text{dc}}(x) + t), \quad t := \frac{b}{|c| \sqrt{d}} \in \mathbb{R} \text{ free.}$$

Orient $\text{sign}(c) = +1$ (otherwise the classes are mislabeled and the margin is negative). The classifier depends on x only through the scalar $f_{\text{dc}}(x)$, so this is a one-dimensional threshold problem on the values $\{f_{\text{dc}}(x) : x \in X_+ \cup X_-\}$. Maximizing $\min_x[\text{margin}]$ over the threshold t gives the symmetric placement $t^* = -\frac{1}{2}(\min_{X_+} f_{\text{dc}} + \max_{X_-} f_{\text{dc}})$ and common value $\frac{1}{2}(\min_{X_+} f_{\text{dc}} - \max_{X_-} f_{\text{dc}})$, which is positive exactly when the projected classes are separated; the $(\cdot)_+$ records the overlap case. Since $\Pi_{\text{inv}}x = f_{\text{dc}}(x) \cdot \frac{1}{\sqrt{d}}\mathbf{1}$ (Lemma 1), this half-gap is the hard-margin of the projected data $\Pi_{\text{inv}}X$. *Status: proved (verified numerically against a 1D SVM and against Ge’s $2/\sqrt{d}$, Section 2.4).* $\square$

Corollary 1 (Invariance never beats the unconstrained margin; the ratio is the diagnostic). *Let $\gamma_{\text{full}} = \text{Sep}(\text{Id} \cdot X)$ be the unconstrained linear max-margin. Then $\gamma_{\text{inv}} \leq \gamma_{\text{full}}$ and $\kappa := \gamma_{\text{inv}}/\gamma_{\text{full}} \in [0, 1]$. Moreover, writing $e = \frac{1}{\sqrt{d}}\mathbf{1}$ and $d_{\text{min}} = \text{dist}(\text{conv } X_+, \text{conv } X_-)$,*

$$\sqrt{d} \left(\min_{X_+} f_{\text{dc}} - \max_{X_-} f_{\text{dc}} \right) = \min_{u \in \text{conv } X_+, v \in \text{conv } X_-} e^\top (u - v) \leq d_{\text{min}},$$

with equality iff the closest opposite pair is parallel to $\mathbf{1}$.

Proof. An invariant linear classifier is a feasible (DC-direction) unconstrained linear classifier, so the constrained max is $\leq$ the unconstrained max. For the inequality, a linear functional over convex hulls attains its min at vertices, and by Cauchy–Schwarz $e^\top (u - v) \leq \|e\| \|u - v\| = \|u - v\|$ with equality iff $u - v \parallel e$. *Status: proved.* $\square$

2.2 The co-existence/trade-off dichotomy and the η/L criterion

Theorem 1 turns “does invariance hurt robustness?” into “does the discriminative signal survive projection onto the invariant feature space?”. If it does, with margin η and the invariant feature having Lipschitz constant L , the invariant classifier is simultaneously fully shift-consistent and robust with radius $\geq \eta/L$. We therefore propose the governing scalar

$$\boxed{\eta/L} \quad (\text{margin of the invariant feature / its Lipschitz constant}).$$

We stress η/L and *not* $\eta > 0$: a signal can survive projection yet have vanishing margin, in which case invariance and robustness still trade off. This is exactly the regime of Kamath et al. [4] (Section 2.4). *Status: the inequality $r_2 \geq \eta/L$ is proved (Corollary 2); “ η/L is the right predictor for trained models” is an empirical claim (Section 4), not a theorem.*

2.3 Nonlinear escape: one convolution-plus-pooling layer computes the power spectrum

The invariant *linear* family is one-dimensional (f_{dc} only, Theorem 1). The nonlinearity in a convolutional layer makes the invariant feature family much larger.

Lemma 2 (Power-spectrum span). *Let $\Phi_w(x) = \frac{1}{d} \sum_{s \in G} ((w \star x)_s)^2$ be a circular cross-correlation with filter w , followed by a quadratic activation and global average pooling. Then*

$$\Phi_w(x) = \frac{1}{d} \sum_{k=0}^{d-1} |\hat{w}_k|^2 |\hat{x}_k|^2,$$

and as w ranges over $\mathbb{R}^d$ the maps $\{\Phi_w\}$ span exactly the power-spectrum features $\{|\hat{x}_k|^2\}_k$, which are shift-invariant; for $k \neq 0$ they are supported on V_{ac} .

Proof. The circular cross-correlation has DFT $(\widehat{w \star x})_k = \widehat{w}_k \hat{x}_k$. By Parseval (unitary DFT), $\frac{1}{d} \sum_s (w \star x)_s^2 = \frac{1}{d} \sum_k |\widehat{w}_k \hat{x}_k|^2 = \frac{1}{d} \sum_k |\widehat{w}_k|^2 |\hat{x}_k|^2$. Choosing $|\widehat{w}_k|^2$ a unit spike at frequency k_0 isolates $|\hat{x}_{k_0}|^2$; linear combinations recover all k . Shift-invariance is Lemma 1. *Status: proved (verified numerically).* (The power spectrum is a *second-order* invariant; it is blind to phase, hence cannot separate sign- or location-coded classes; recovering phase needs the bispectrum, whose Lipschitz constant we do not control.) $\square$

Corollary 2 (Conditional robust radius). *Let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be a shift-invariant feature map that is L -Lipschitz on a neighbourhood of the data, and let a unit-norm linear head separate $\Psi(X_+), \Psi(X_-)$ with margin $\eta := \frac{1}{2} \text{dist}(\text{conv } \Psi(X_+), \text{conv } \Psi(X_-))$. Then the classifier $g = \langle a, \Psi(\cdot) \rangle + \beta$ has SC = 1 and $r_2(g, x) \geq \eta/L$ for every data point.*

Proof. Invariance of Ψ gives $g(S_s x) = g(x)$, so SC = 1. For the radius, $|g(x + \delta) - g(x)| = |\langle a, \Psi(x + \delta) - \Psi(x) \rangle| \leq \|a\| L \|\delta\| = L \|\delta\|$; if $\|\delta\| < \eta/L$ this is $< \eta \leq y g(x)$, so the sign cannot change. *Status: proved, but it is the standard Lipschitz-margin bound; its content is which Ψ a shift-invariant architecture can realize (Lemma 2), not the inequality. We flag the novelty question in Section 6.* $\square$

Lemma 3 (Orbit-gradient suppression, continuous group). *Let $\{T_a\}_{a \in \mathbb{R}}$ be the continuous circular-shift group with generator $Gx := \frac{d}{da} T_a x|_{a=0}$. If g is invariant under $\{T_a\}$ and differentiable, then $\langle \nabla g(x), Gx \rangle = 0$ for all x : the input gradient is orthogonal to the orbit tangent, so first-order perturbations along the orbit do not change the score.*

Proof. Differentiate $g(T_a x) = g(x)$ at $a = 0$: $\langle \nabla g(x), Gx \rangle = 0$. *Status: proved (verified, $\approx 10^{-9}$). Caveat: this is the continuous orbit tangent. The discrete chord $S_1 x - x$ is a finite secant, not a tangent, and is in general not protected ($\langle \nabla g, S_1 x - x \rangle \neq 0$); for the discrete threat model the protection is only first-order along the continuous orbit.* $\square$

2.4 Recoveries and reconciliations

Ge et al. [3] (recovered exactly). For the single white/black dot, $X_+ = \mathcal{O}(e_j)$, $X_- = \mathcal{O}(-e_j)$: $f_{dc}(\pm e_j) = \pm 1/\sqrt{d}$, so (1) gives $\gamma_{\text{inv}} = 1/\sqrt{d}$; in Ge’s one-sided convention the gap is $2/\sqrt{d}$ (their Cor. 1), while the unconstrained margin is 1, so $\kappa = 1/\sqrt{d} \rightarrow 0$ (the forced-trade-off corner). The power spectrum is sign-blind, $|\widehat{+e_j}_k|^2 = |\widehat{-e_j}_k|^2$, so $\eta_{\text{PS}} = 0$: even the nonlinear invariant cannot separate the dot. *Status: proved/verified.*

Kamath et al. [4] (corrected reconciliation). Their data places the label on a coordinate x_0 that their shift *fixes*, i.e. $x_0 \in V_{\text{inv}}$. Hence an invariant classifier reads x_0 and attains high clean accuracy (their Prop. 1, $\geq 97\%$), so $\eta > 0$. The trade-off (their Thm. 2) arises because x_0 has ℓ_∞ margin of order $1/\sqrt{d}$: this is the *small- η/L* corner of our criterion, not a case where invariance discards the signal. Their impossibility is therefore an *instance* of the criterion, not a counterexample. *Status: verified against their construction (read firsthand).* **This corrects an earlier draft that claimed Kamath’s signal lives outside V_{inv} ; it does not.**

Tramèr et al. [7] (recovered as a corollary). They show that *too much* invariance creates “invariance-based” adversarial examples: a perturbation that changes the oracle (true) class while the model keeps its prediction. In our language this is exactly the regime where the orbit of the imposed invariance crosses an oracle decision boundary, i.e. where projecting onto V_{inv} (or onto Ψ) collapses the class signal ($\eta \rightarrow 0$, equivalently $\kappa \rightarrow 0$). Their negative result is the same forced-trade-off corner;

co-existence is the complementary $\eta/L = \Theta(1)$ regime. *Status: argued, not formalized into a single statement; see Section 6, item 5.*

An illustrative nonlinear case (toy). Let each class be the orbit (phases) of a single cosine at distinct frequencies $k_+ \neq k_-$, amplitude a . The invariant score $f(x) = |\hat{x}_{k_+}|^2 - |\hat{x}_{k_-}|^2$ separates with exact ℓ_2 robust radius $a/\sqrt{2}$ (minimize $|p|^2 + |q|^2$ s.t. $|a + p|^2 \leq |q|^2$; optimum $p = -a/2$), while any *linear* classifier has margin 0 (each class is a circle through the origin in an orthogonal 2-plane). So invariance strictly helps. *Status: exact for this toy; **not** the quantitative statement for Ge et al. [3]’s Section 5.2 data, whose classes are spans of many frequencies (clouds in power-spectrum space), where the qualitative reversal survives but the exact $a/\sqrt{2}$ does not. Confirmed training-free: on the multi-frequency span data, $\text{Sep}(\text{Id}) = \text{Sep}(\Pi_{\text{inv}}) = 0$ but the power-spectrum margin is positive (Section 4).*

3 Optimization: does gradient descent reach the robust invariant solution?

Theorem 1 and Corollary 2 say robust shift-invariant classifiers *exist*. They do not say gradient descent *finds* them. This is the same existence-vs-trajectory gap as Frei et al. [1] (robust two-layer ReLU nets exist, but gradient flow selects a non-robust max-margin KKT point) and Li et al. [5] (the non-robust selection is *feature averaging*); Min and Vidal [6] conjecture that an architectural change re-points the bias.

Lemma 4 (Orbit-bundle form). *A circular-convolution-plus-global-average-pooling network equals a two-layer network whose hidden neurons are tied into shift-orbit bundles (one filter induces all d of its shifts, sharing a single output weight), and whose pre-head feature is an average over the orbit.* Status: proved (exact, structural; verified $\approx 10^{-9}$).

Retraction. An earlier draft claimed that this orbit-bundle structure *converifies* the margin program and *annihilates* the feature-averaging direction of Frei et al. [1], Li et al. [5], thereby turning the Min and Vidal [6] conjecture into a theorem for the quadratic head. **This is wrong and is retracted.** The error, shared by four independent derivations, is the next proposition.

Proposition 1 (Single-orbit data is a degenerate corner of the implicit-bias models). *Let each class be the shift orbit of a single base pattern.*

1. *If the base is a cosine at frequency k , its orbit (all phases) spans only $\{\cos, \sin\}$ at frequency k , so the per-class data has **rank** 2, violating the near-orthogonal $\Omega(d)$ -cluster assumptions of Frei et al. [1], Li et al. [5]. Moreover the signed mean of the orbit, which is the direction carrying the non-robustness conclusion in those works, is **0**.*
2. *If the base is a localized spike e_j , the orbit is orthonormal (so those assumptions hold), but the power spectrum is phase-blind, so $\eta_{\text{PS}} = 0$ and the invariant nonlinear classifier cannot separate the two classes.*

Hence no single orbit base pattern simultaneously satisfies the non-robustness premise of Frei et al. [1], Li et al. [5] and admits an invariant-feature separation: the contrast “unconstrained net non-robust, invariant net robust, along the same threat direction” is self-cancelling.

Proof. (i) All circular shifts of $\cos(2\pi k \cdot /d)$ are linear combinations of $\cos(2\pi k \cdot /d)$ and $\sin(2\pi k \cdot /d)$ (shift is a phase rotation of a single frequency), so the orbit lies in a 2-dimensional space; numerically

the orbit matrix for $d = 64, k = 5$ has rank 2. The orbit mean is $\frac{1}{d} \sum_s S_s u = (\frac{1}{d} \mathbf{1} \mathbf{1}^\top) u = f_{\text{dc}}(u) \frac{1}{\sqrt{d}} \mathbf{1} = \mathbf{0}$ for a zero-mean cosine; numerically $\|\text{mean}\| = 1.7 \times 10^{-16}$. (ii) Shifts of e_j are the standard basis, orthonormal; $|\hat{e}_{j_k}|^2 = 1/d$ for all k independent of j and of sign, so the two classes coincide in power-spectrum space, $\eta_{\text{PS}} = 0$ (numerically the power-spectrum gap is 0). *Status: proved and verified numerically.* $\square$

Remark 1 (What survives, and why the failure is informative). Proposition 1 is a correct, modest *structural* observation: single-orbit-per-class data sits in a rank-deficient degenerate corner of the cluster models used to prove implicit-bias non-robustness, a corner where the non-robustness mechanism has no content. This explains both why an unconstrained network is already competitive on such data and why a naive “invariance re-points the bias” argument dissolves. It does *not* close the optimization gap; it tells us the gap must be studied on data where the non-robustness premise genuinely holds (Section 4, Direction 1).

4 Open problems, conjectures, and directions (with empirical hints)

Empirical hints currently available.

- **Training-free structural test (confirmed).** With $d = 64$: the dot has $\text{Sep}(\text{Id}) = 1$, $\text{Sep}(\Pi_{\text{inv}}) = 1/\sqrt{d} = 0.125$, $\eta_{\text{PS}} = 0$; the orthogonal-frequency *span* data of Ge et al. [3] has $\text{Sep}(\text{Id}) = \text{Sep}(\Pi_{\text{inv}}) = 0$ but power-spectrum margin > 0 . So the right invariant feature, not invariance per se, preserves the margin.
- **MNIST architectural dissection (suggestive, under cross-dataset re-test).** Across seven architectures, η/L (estimated as mean logit margin over mean input-gradient norm) tracked ℓ_2 robustness with Pearson $r \approx 0.97$, while shift-consistency did not ($r \approx -0.29$); the fully shift-invariant single-layer model had consistency 1 but was the least robust. Batch-normalization raised clean accuracy and lowered robustness (consistent with Galloway et al. [2]). *Caveat: on Fashion-MNIST the fixed- ϵ version did not replicate (robust accuracies near the floor; metric artifact); a dataset-comparable robust-radius re-test is in progress. The cross-dataset status of “ η/L predicts robustness” is therefore currently open.*

Open question 1 (Cross-dataset validity of the η/L predictor). Is “ η/L predicts the robust radius of trained models, better than shift-consistency” a robust empirical law across datasets, attacks (PGD, AutoAttack), and capacity-matched architectures? If it is dataset-dependent, which data property (intrinsic dimension, frequency content, margin distribution) decides it? A natural decomposition to measure: does an architecture’s robustness advantage come from larger margin η , smaller Lipschitz L , or both?

Conjecture 1 (Multi-orbit implicit bias — the salvageable optimization claim). *Take data where each class is a union of many orbits of distinct base patterns, so the data is high-rank (the Frei et al. [1], Li et al. [5] premise can hold) and carries genuine shift structure. Conjecture: in this regime the orbit-bundle constraint (Lemma 4) makes gradient flow average over the within-orbit (nuisance) directions while an unconstrained network averages over class-discriminative directions, so the convolutional network reaches a strictly larger η/L . This is the version of the retracted idea that is not self-cancelling, because the rank-deficiency of Proposition 1 no longer holds.* Status: conjecture; the right next theorem to attempt.

Open question 2 (Implicit bias of the orbit-bundle constraint). Analyse the max-margin / KKT limit (Lyu–Li, Ji–Telgarsky) of the weight-shared conv-plus-pool network on the multi-orbit data

of Conjecture 1. Does weight-sharing change the *reachable* set of KKT points relative to the unconstrained net, in the direction of higher margin? This is precisely where Min and Vidal [6] leave a conjecture (for the activation lever); the shift constraint is a different, deployable lever.

Open question 3 (Is η/L a genuinely new robustness quantity?). Corollary 2 is a standard Lipschitz–margin bound specialized to invariant features. What is the novel content beyond (a) Theorem 1’s exact projection identity and (b) the architectural realizability of the power spectrum (Lemma 2)? Is there a converse (an upper bound on r_2 in terms of η/L) that would make η/L a characterization rather than a one-sided proxy?

Open question 4 (Beyond cyclic shifts). Does the criterion generalize to other compact groups (rotations, scalings; the equivariance setting of Kamath et al. [4] and follow-ups)? The invariant subspace generalizes by Peter–Weyl; the open part is the analogue of the power-spectrum escape (which invariant nonlinear features a given equivariant architecture realizes) and whether η/L still governs.

Open question 5 (Certified robustness). η/L is a *certified* lower bound (Corollary 2). Can a shift-invariant architecture with a controlled-Lipschitz invariant feature give certified robustness competitive with empirical AutoAttack robustness, and does the certificate explain the Galloway batch-norm effect through the L term?

Open question 6 (The Tramèr trade-off, formalized). Make the recovery of Tramèr et al. [7] a single statement: an inequality whose two sides are the sensitivity-based robust radius and an invariance-based (oracle-flipping) radius, with η/L (and the orbit-to-oracle-boundary distance) as the control, so that co-existence and the invariance-based attack are the two regimes of one bound.

5 Overclaim audit (status of every claim)

Claim	Status
Lemma 1 ($V_{\text{inv}} = \text{span}\mathbf{1}$, DFT diagonalization)	proved
Theorem 1 (invariant linear margin = projection separation)	proved + verified
Corollary 1 ($\kappa \in [0, 1]$; $\sqrt{d} \Delta_{\text{DC}} \leq d_{\text{min}}$)	proved
Lemma 2 (conv+sq+GAP spans the power spectrum)	proved + verified
Corollary 2 ($r_2 \geq \eta/L$)	proved; standard Lipschitz–margin; conditional on η ,
Lemma 3 (orbit-gradient = 0)	proved for the <i>continuous</i> group only
Ge $2/\sqrt{d}$ recovery	proved + verified
Kamath = small- η/L corner	verified against the source
Tramèr = forced-trade-off corner	argued, not yet a single formal statement
Toy $a/\sqrt{2}$ nonlinear radius	exact for the single-frequency toy only; <i>not</i> Ge’s span
“ η/L predicts trained-model robustness”	empirical: MNIST yes, cross-dataset <i>open</i>
“architecture re-points the implicit bias / Min&Vidal becomes a theorem”	RETRACTED
Proposition 1 (single-orbit data is a degenerate corner)	proved + verified
Conjecture 1 (multi-orbit implicit bias)	conjecture (open)
A trajectory-level implicit-bias theorem for shift-invariant nets	open

6 What I would most like a reviewer to check or prove

1. **Theorem 1 proof**, in particular the step “invariance $\Leftrightarrow w \in \text{span}(\mathbf{1})$ ” and the one-dimensional margin maximization; and whether the convex-hull form (used in Corollary 2) should replace

$\text{Sep}(\Pi_{\text{inv}}X)$ everywhere (they coincide only for single-orbit-per-class data).

2. **Lemma 2** normalization (the unitary-DFT factor $1/d$) and the “spans exactly” claim.
3. **Kamath reconciliation:** confirm against Kamath et al. [4] that x_0 is shift-fixed and has ℓ_∞ margin $O(1/\sqrt{d})$, so their impossibility is our small- η/L corner.
4. **Novelty of η/L** (the question in Section 4): is Corollary 2 more than a specialized Lipschitz-margin bound? Is there a matching converse? This most affects whether the structural part is a “contribution” or a “clarification”.
5. **Tramèr formalization** (Section 4): is the proposed single inequality correct and provable?
6. **Conjecture 1** (the open optimization theorem): is the multi-orbit setting the right one, is it tractable, and is there prior work (equivariant implicit bias) that already settles part of it?
7. **Discrete vs continuous orbit-gradient** (Lemma 3): does the continuous-tangent protection matter for the discrete (integer-shift) threat model, or is it cosmetic?

Honest assessment of where this stands. The structural theory is correct and clean but, on its own, is close to “recover Ge et al. [3] and state a margin/Lipschitz criterion”. To be a strong paper rather than an exercise it needs at least one of: (a) the optimization theorem of Conjecture 1; (b) a robust, cross-dataset, AutoAttack-validated empirical law for η/L (Open question 1) with a capacity-matched dissection; or (c) the certified-robustness or general-group extension. These are the levers; the experiments in progress target (b).

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